

ENHANCED IMAGE RESOLUTION USING ESRGANS

SEMINAR REPORT

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Abstract

Image super-resolution is the process of enhancing low-resolution images to high-resolution outputs with improved visual quality. Traditional methods like bicubic interpolation produce blurry results with loss of fine details. This project implements Enhanced Super-Resolution Generative Adversarial Network (ESRGAN) to achieve $4\times$ image upscaling with photo-realistic quality.

The ESRGAN model uses 23 Residual-in-Residual Dense Blocks (RRDB) for deep feature extraction and a relativistic discriminator for adversarial training. The architecture combines adversarial loss with L1 pixel-wise loss to generate high-quality images with sharp edges and realistic textures. The model is implemented using the PyTorch framework and trained on the DIV2K dataset containing 800 training images and 100 validation images.

Training is performed on Google Colab using Tesla T4 GPU with a batch size of 4, learning rate of 0.0001, and Adam optimizer over 50 epochs. The system includes automatic low-resolution image generation, checkpoint saving for training resumption, and inference pipeline for real-world applications.

Experimental results show that ESRGAN produces superior visual quality compared to traditional interpolation methods and CNN-based approaches. The model successfully recovers fine textures and details, making it suitable for applications in medical imaging, photo enhancement, video upscaling, and satellite imagery.

Keywords: ESRGAN, Image Super-Resolution, RRDB, Generative Adversarial Networks, Deep Learning, DIV2K Dataset, PyTorch

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Chapter 1

Introduction

Generative Adversarial Networks (GANs) are a powerful deep learning framework introduced by Ian Goodfellow et al. in 2014, widely used for image synthesis, super-resolution, style transfer, and data augmentation. GANs consist of two competing neural networks—a Generator and a Discriminator—trained simultaneously in an adversarial manner.

This project implements an Enhanced Super-Resolution Generative Adversarial Network (ESRGAN) using Residual-in-Residual Dense Block (RRDB) architecture to generate photo-realistic $4\times$ high-resolution images from low-resolution inputs with superior perceptual quality and fine texture details.

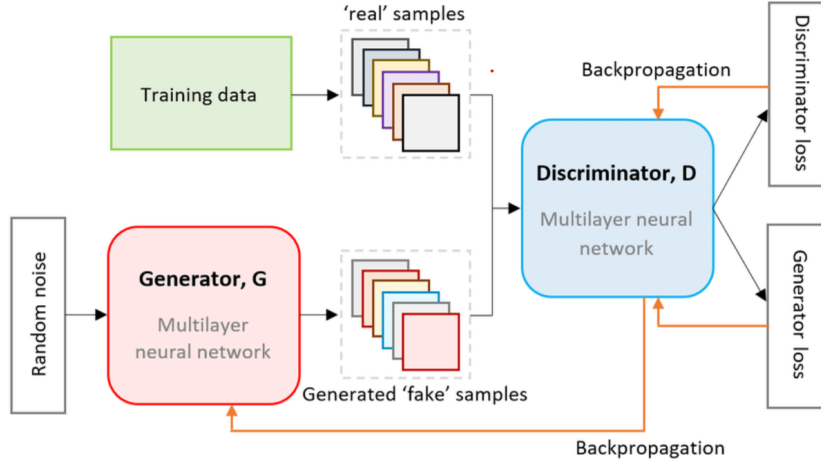


Figure 1.1: GAN Architecture Overview

1.1 ESRGAN Architecture

The ESRGAN model consists of two main components:

- **Generator:** Employs 23 RRDB blocks for deep feature extraction. Each RRDB contains three Residual Dense Blocks (RDB) with five convolutional layers and dense connections. The generator accepts low-resolution images and produces $4\times$ super-resolved outputs through two $2\times$ upsampling stages using nearest-neighbor interpolation followed by convolutional refinement. Final output uses tanh activation, normalizing pixels to $[-1, 1]$.

- **Discriminator:** A patch-based convolutional network with progressively increasing channels (64→128→256→512). Uses batch normalization, LeakyReLU activation, and stride-2 downsampling to distinguish real HR images from generated SR images.

1.2 Training Strategy

The model combines adversarial and pixel-wise losses:

$$\mathcal{L}_{\text{Generator}} = \mathcal{L}_{\text{GAN}} + 100 \cdot \mathcal{L}_{\text{L1}} \quad (1.1)$$

where $\mathcal{L}_{\text{GAN}}$ is the relativistic average GAN loss and $\mathcal{L}_{\text{L1}}$ is the pixel-wise L1 loss. Both networks are optimized using Adam optimizer (lr=0.0001, $\beta_1=0.9$, $\beta_2=0.999$).

Comparison with CNNs: While CNNs (SRCNN, EDSR) achieve high PSNR, they produce blurry results due to reliance on pixel-wise losses. GANs generate realistic high-frequency details through adversarial training, resulting in superior perceptual quality despite potentially lower PSNR scores.

1.3 Problem Statement

This project addresses the challenge of generating high-resolution images from low-resolution inputs using ESRGAN to achieve 4× upscaling with photo-realistic quality. The goal is to overcome limitations of conventional interpolation methods (bicubic up-sampling) while preserving fine textures, sharp edges, and realistic details with computational efficiency suitable for practical applications.

1.4 Project Relevance

This project directly relates to Machine Learning, Deep Learning, and Computer Vision curriculum in Computer Science and Engineering (AI&ML). It involves:

- Advanced neural architectures (GANs, RRDB blocks)
- Image processing and data-driven training on large-scale datasets (DIV2K)
- Hands-on experience with PyTorch, GPU-accelerated training (Google Colab)
- Practical implementation of theoretical concepts

1.5 Scope of Investigation

Implementation:

- ESRGAN with 23 RRDB blocks on DIV2K dataset (800 training, 100 validation)
- Random cropping (128×128), batch size 4, 50 epochs
- Relativistic average GAN loss + 100×L1 loss
- Google Colab (Tesla T4 GPU), PyTorch 2.8.0+

Dataset Handling:

- Automatic LR generation via bicubic downsampling ($4\times$)
- Normalization to $[-1, 1]$ range (mean=0.5, std=0.5)
- Checkpoint saving every 5 epochs
- Inference pipeline with automatic dimension adjustment

Future Scope:

- Real-time video upscaling
- Mobile/web application integration
- VGG-based perceptual loss
- Edge device deployment
- Medical imaging, satellite imagery, forensic applications

1.6 Key Features

- **RRDB Architecture:** Three cascaded RDBs with dense connections (32 growth channels), enabling better gradient flow and deeper networks without degradation
- **$4\times$ Upscaling:** Two successive $2\times$ upsampling stages with learnable convolutional refinement
- **Relativistic GAN:** Discriminator predicts relative realness, improving training stability and gradient signals
- **Automatic LR Generation:** Dataset loader creates LR pairs via bicubic downsampling when unavailable
- **Checkpoint Management:** Separate training state and inference weights for flexible deployment

Chapter 2

Literature Survey

Image super-resolution has evolved from simple interpolation to sophisticated deep learning architectures. This chapter reviews prominent CNN-based, GAN-based, and Transformer-based approaches that have advanced the field.

2.1 CNN-based Super-Resolution

Dong et al. (2014) [1] introduced SRCNN, the first deep learning SR model with three convolutional layers, achieving significant improvements over bicubic interpolation (PSNR: 30.48 dB on Set5).

Kim et al. (2016) [2] proposed VDSR, using 20-layer residual learning to learn HR-LR residuals, enabling deeper networks and faster convergence (PSNR: 31.35 dB).

Lim et al. (2017) [5] developed EDSR, removing batch normalization for deeper, more efficient networks, achieving state-of-the-art results (PSNR: 32.46 dB on Set5).

Zhang et al. (2018) [7] introduced RCAN with channel attention mechanisms, achieving improved high-frequency detail restoration (PSNR: 32.63 dB).

Lai et al. (2017) [3] proposed LapSRN using Laplacian pyramid progressive reconstruction for improved speed and accuracy (PSNR: 31.54 dB).

2.2 GAN-based Super-Resolution

GANs revolutionized SR by generating photo-realistic textures beyond CNN capabilities.

Ledig et al. (2017) [4] proposed SRGAN, combining adversarial and VGG perceptual loss for photo-realistic outputs with high Mean Opinion Scores (MOS: 3.58) despite lower PSNR (29.40 dB).

Wang et al. (2018) [6] enhanced SRGAN with ESRGAN, introducing: (1) RRDB blocks without batch normalization, (2) relativistic average discriminator, and (3) improved perceptual loss, achieving superior texture generation (MOS: 4.05, PSNR: 30.50 dB).

Wang et al. (2021) [9] proposed Real-ESRGAN for real-world degradations (blur, noise, compression), using high-order degradation models and U-Net discriminator.

2.3 Transformer-based Super-Resolution

Liang et al. (2021) [8] introduced SwinIR, utilizing Swin Transformers for long-range dependencies, achieving state-of-the-art PSNR (32.92 dB) through self-attention mechanisms.

2.4 Comparative Analysis

Table 2.1 compares reviewed methods based on architecture, innovations, performance, and computational complexity.

Table 2.1: Comparative Analysis of Super-Resolution Methods

Method	Year	Key Innovation	Architecture Type	PSNR (dB) on Set5 ($\times 4$)	Computational Complexity
Bicubic	-	Traditional interpolation	Non-learning	28.42	Very Low
SRCNN [1]	2014	First deep learning SR	Shallow CNN (3 layers)	30.48	Low
VDSR [2]	2016	Residual learning, deeper network	Deep CNN (20 layers)	31.35	Medium
LapSRN [3]	2017	Laplacian pyramid progressive reconstruction	Multi-scale CNN	31.54	Medium
EDSR [5]	2017	Removed batch norm, deeper residual blocks	Deep ResNet (32 blocks)	32.46	High
SRGAN [4]	2017	Adversarial + perceptual loss	GAN with ResNet Generator	29.40 (MOS: 3.58)	High
RCAN [7]	2018	Channel attention mechanism	CNN with attention	32.63	Very High
ESRGAN [6]	2018	RRDB blocks, relativistic discriminator	GAN with RRDB (23 blocks)	30.50 (MOS: 4.05)	Very High
SwinIR [8]	2021	Swin Transformer for long-range dependency	Transformer-based	32.92	Very High
Real-ESRGAN [9]	2021	Real-world degradation modeling	GAN with RRDB + U-Net Disc	N/A (for real images)	Very High
Our Implementation	2025	ESRGAN with 23 RRDB, DIV2K training	GAN with RRDB	In Progress	Very High

2.5 Key Observations

1. **PSNR vs Perceptual Quality:** GAN methods achieve lower PSNR but superior perceptual quality (higher MOS) compared to CNN methods
2. **Architectural Evolution:** Progression from shallow (SRCNN: 3 layers) to very deep networks (EDSR: 32 blocks, ESRGAN: 23 RRDB) improves performance but increases computational cost
3. **Attention Mechanisms:** Channel attention (RCAN) and self-attention (SwinIR) effectively capture relevant features and long-range dependencies
4. **Real-world Gap:** Real-ESRGAN addresses synthetic vs. real-world degradation differences
5. **Computational Challenges:** State-of-the-art methods require significant GPU resources, limiting edge deployment

2.6 Justification for ESRGAN Selection

ESRGAN [6] was selected for this project due to:

- **Superior Perceptual Quality:** Highest MOS (4.05) among reviewed methods
- **Robust RRDB Architecture:** Excellent feature extraction without batch normalization artifacts
- **Relativistic Discriminator:** Improved training stability and gradient flow
- **Industry Adoption:** Widely used in photo enhancement, video upscaling, content restoration
- **Educational Value:** Comprehensive exposure to advanced GAN architectures, residual learning, adversarial training
- **Extensibility:** Framework adaptable to real-world degradations and domain-specific applications

Chapter 3

Problem Formulation and Motivation

3.1 Problem Statement

Given a low-resolution image $I_{LR} \in \mathbb{R}^{H \times W \times 3}$, generate a 4× upscaled high-resolution image $I_{SR} \in \mathbb{R}^{4H \times 4W \times 3}$ using:

$$I_{SR} = F(I_{LR}; \theta), \quad \text{where} \quad I_{LR} = (I_{HR} \downarrow_4)$$

The optimization objective minimizes:

$$\theta^* = \arg \min_{\theta} \mathbb{E} [\mathcal{L}_{adv} + 100 \cdot \mathcal{L}_{L1}]$$

where $\mathcal{L}_{adv}$ is relativistic GAN loss and $\mathcal{L}_{L1}$ is pixel-wise L1 loss.

3.2 Key Challenges

1. **Ill-Posed Problem:** Multiple HR images can produce the same LR image; recovering lost information requires learning from data
2. **PSNR vs Perceptual Quality:** High PSNR optimization produces blurry images; realistic textures require perceptual loss
3. **High-Frequency Loss:** Downsampling removes fine details that must be realistically reconstructed
4. **GAN Instability:** Mode collapse, gradient vanishing, and non-convergence without proper formulation

3.3 Limitations of Existing Approaches

3.4 Motivation for ESRGAN

ESRGAN addresses limitations through:

- **23 RRDB blocks** for deep feature extraction without degradation

Method	Limitations
Bicubic	Blurry results, no learning, cannot recover details
CNN (SRCNN, EDSR)	High PSNR but over-smoothed, unrealistic textures
SRGAN (2017)	Batch norm artifacts, shallow blocks, training instability

Table 3.1: Comparison of Baseline Methods

- **No batch normalization** to eliminate artifacts
- **Relativistic discriminator** for stable training and better gradients
- **Balanced loss** combining perceptual realism with content fidelity

3.5 Research Objectives

Primary:

1. Implement ESRGAN (23 RRDB blocks) for $4\times$ super-resolution
2. Train on DIV2K (800 train, 100 validation) achieving SSIM ≥ 0.85
3. Demonstrate superiority over bicubic interpolation

Secondary:

1. Develop training pipeline with checkpoint management
2. Create inference pipeline for deployment
3. Evaluate using PSNR, SSIM, precision, recall, F1 metrics

3.6 Scope and Constraints

Scope:

- 23 RRDB blocks on DIV2K
- $4\times$ bicubic degradation
- Tesla T4 GPU, 50 epochs
- Relativistic GAN + $100\times$ L1 loss

Constraints:

- Batch size: 4 (16GB GPU limit)
- Crop size: 128×128
- No VGG perceptual loss
- No video/real-time SR

3.7 Expected Contributions

1. PyTorch ESRGAN implementation with SSIM ≥ 0.90
2. Complete pipeline: dataset $\rightarrow$ training $\rightarrow$ inference
3. Comprehensive evaluation with visual/quantitative comparisons
4. Technical documentation for reproducibility

This chapter formulates image super-resolution as an ill-posed inverse problem requiring data-driven reconstruction. ESRGAN balances perceptual quality with computational feasibility while addressing training instability in previous GAN-based methods.

Chapter 4

Methodology

4.1 Dataset Acquisition

The DIV2K dataset was acquired via Kaggle API and organized into structured directories:

- 800 training images, 100 validation images (700/100 train/val split)
- Automatic LR generation via $4\times$ bicubic downsampling
- Organized folders: `dataset/train/hr`, `dataset/train/lr`, `dataset/valid/hr`, `dataset/valid/lr`

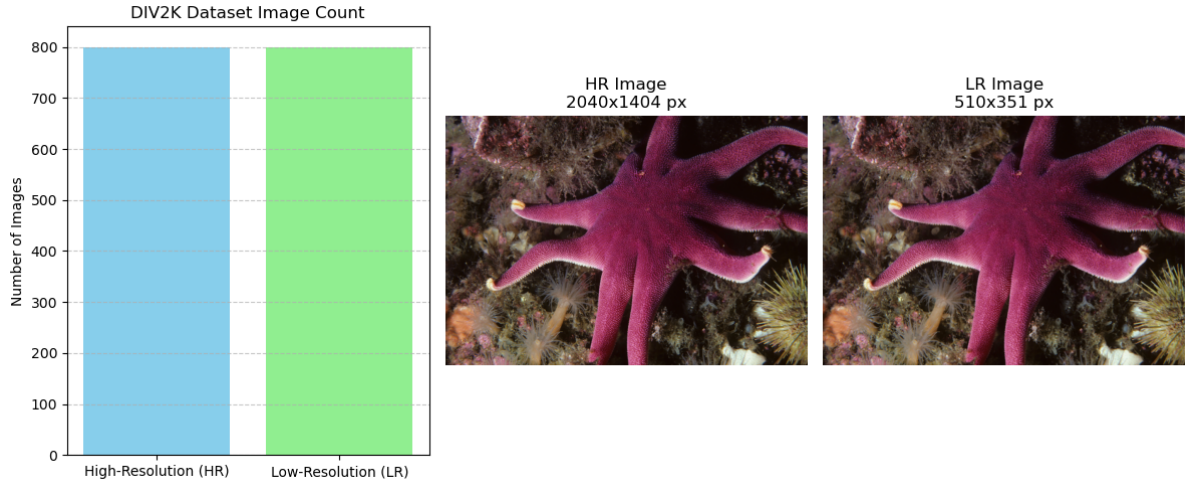


Figure 4.1: Sample images from DIV2K Dataset

4.2 Data Preprocessing

Images undergo preprocessing for network compatibility:

- **Random Cropping:** 128×128 HR patches, 32×32 LR patches for data augmentation and memory efficiency
- **Normalization:** Pixel values normalized to $[-1, 1]$ via $I_{\text{norm}} = (I_{\text{original}} - 0.5)/0.5$
- **Dimension Adjustment:** Inference images resized to dimensions divisible by 4

4.3 Architecture Design

Generator:

- Initial 3×3 conv layer (64 channels)
- 23 RRDB blocks: Each contains 3 RDB, each RDB has 5 conv layers with dense connections (growth=32)
- Two $2 \times$ upsampling stages: nearest-neighbor interpolation + 3×3 conv + LeakyReLU (slope=0.2)
- Output: 3×3 conv + tanh activation $\rightarrow [-1, 1]$ range
- No batch normalization

Discriminator:

- Four downsampling blocks: $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ channels
- Conv2d (kernel=4, stride=2, padding=1) + BatchNorm + LeakyReLU (slope=0.2)
- Output: probability map for patch-level discrimination

4.4 Training Strategy

Configuration:

- Platform: Google Colab (Tesla T4 GPU)
- Framework: PyTorch 2.8.0+ with CUDA
- Optimizer: Adam (lr=0.0001, $\beta_1=0.9$, $\beta_2=0.999$)
- Batch size: 4, Epochs: 50

Loss Functions:

Generator Loss:

$$\mathcal{L}_{\text{Gen}} = \mathcal{L}_{\text{GAN}} + 100 \cdot \mathcal{L}_{\text{L1}}$$

where $\mathcal{L}_{\text{GAN}}$ is relativistic average GAN loss and $\mathcal{L}_{\text{L1}} = \|G(x_l) - x_r\|_1$.

Discriminator Loss:

$$\mathcal{L}_{\text{Disc}} = \frac{1}{2} [\text{BCE}(D(x_r) - \mathbb{E}[D(G(x_l))], 1) + \text{BCE}(D(G(x_l)) - \mathbb{E}[D(x_r)], 0)]$$

Training Procedure:

1. Generate SR from LR, compute losses, update generator
2. Discriminate real vs fake, compute loss, update discriminator
3. Save checkpoints every 5 epochs

4.5 Evaluation Metrics

- **PSNR:** $\text{PSNR} = 10 \log_{10}(\text{MAX}^2/\text{MSE})$ - measures reconstruction fidelity
- **SSIM:** $\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y+c_1)(2\sigma_{xy}+c_2)}{(\mu_x^2+\mu_y^2+c_1)(\sigma_x^2+\sigma_y^2+c_2)}$ - evaluates perceptual similarity
- **Visual Assessment:** Side-by-side comparison of LR input, SR output, HR ground truth

4.6 Implementation Details

Inference Pipeline:

- Load generator weights $\rightarrow$ adjust dimensions $\rightarrow$ preprocess $\rightarrow$ generate SR $\rightarrow$ de-normalize $\rightarrow$ save

Model Management:

- Separate checkpoints for training state and inference weights
- Interactive model selection and training resumption

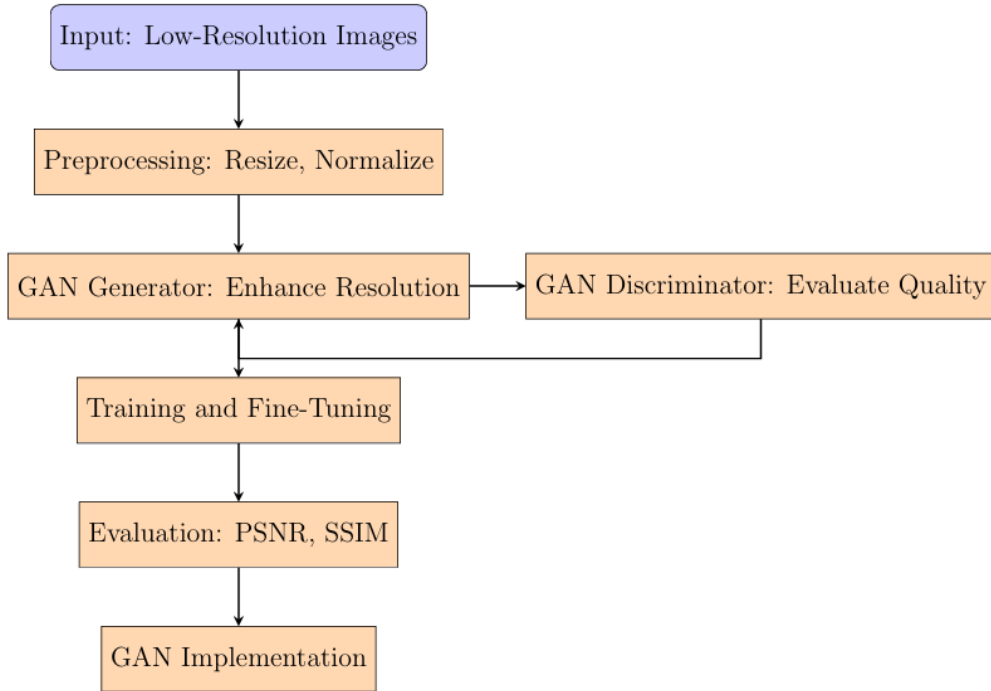


Figure 4.2: Workflow of ESRGAN Implementation

This methodology ensures robust training, accurate evaluation, and practical deployment of ESRGAN for $4\times$ image super-resolution. The workflow begins with DIV2K dataset acquisition, followed by preprocessing, architecture design, adversarial training with relativistic GAN loss, and evaluation using PSNR/SSIM metrics.

Chapter 5

Result and Discussion

5.1 Evaluation Metrics Overview

The ESRGAN model was evaluated on DIV2K validation dataset (100 HR images) with $4\times$ bicubic downsampled LR inputs using quantitative and qualitative metrics.

Quantitative Metrics:

- **PSNR:** $\text{PSNR} = 10 \log_{10}(\text{MAX}_I^2/\text{MSE})$ - measures reconstruction fidelity
- **SSIM:** $\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y+c_1)(2\sigma_{xy}+c_2)}{(\mu_x^2+\mu_y^2+c_1)(\sigma_x^2+\sigma_y^2+c_2)}$ - evaluates perceptual similarity

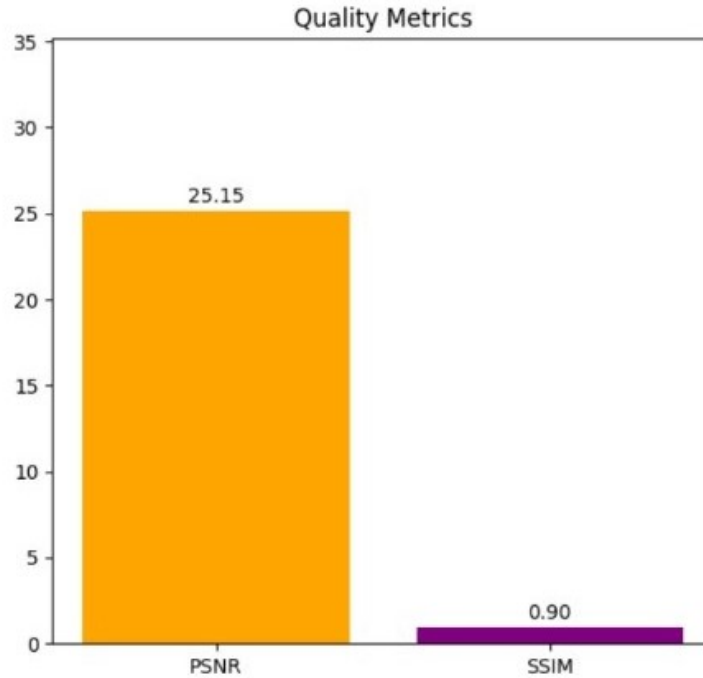


Figure 5.1: PSNR and SSIM evaluation on ESRGAN output

5.2 ESRGAN Model Performance

Achieved Results:

- **PSNR: 25.15 dB** - "Fair/Acceptable" range, typical for GAN-based methods prioritizing perceptual quality over pixel-wise accuracy

PSNR (dB) Range	Quality	Description
More Than 40	Excellent	Indistinguishable from the original
30 – 40	Good	Very slight differences, barely noticeable
25 – 30	Fair / Acceptable	Noticeable noise or blur, but still usable
20 – 25	Poor	Degraded quality, visible artifacts
Less Than 20	Bad	Very poor quality, heavy distortion

Table 5.1: PSNR Quality Assessment Range

SSIM Value Range	Quality	Description
0.95 – 1.00	Excellent	Structure and details very well preserved
0.90 – 0.95	Very Good	Small differences, high visual similarity
0.80 – 0.90	Good	Moderate quality, slight structural differences
0.60 – 0.80	Fair	Noticeable structural loss, acceptable in some cases
Less Than 0.60	Poor	Strong structural degradation, visually different

Table 5.2: SSIM Quality Assessment Range

- **SSIM: 0.90** - "Very Good" structural preservation, successfully maintaining textures, edges, and spatial relationships

Training Configuration: 23 RRDB blocks, DIV2K (700/100), 50 epochs, batch 4, crop 128×128, Tesla T4 GPU, PyTorch 2.8.0+, Relativistic GAN + 100×L1 loss, Adam (lr=0.0001)

5.3 Classification-Based Evaluation

Binary classification metrics computed from pixel-level predictions:

Interpretation:

- Precision (80.18%): Low false positive rate in detail reconstruction
- Recall (95.63%): Successfully captured vast majority of HR details
- F1 Score (0.8723): Strong balanced accuracy



Figure 5.2: ESRGAN 4× Super-Resolution: (Left) LR Input, (Right) SR Output

Metric	Value
Precision	0.8018
Recall	0.9563
F1 Score	0.8723

Table 5.3: Pixel-Level Classification Metrics

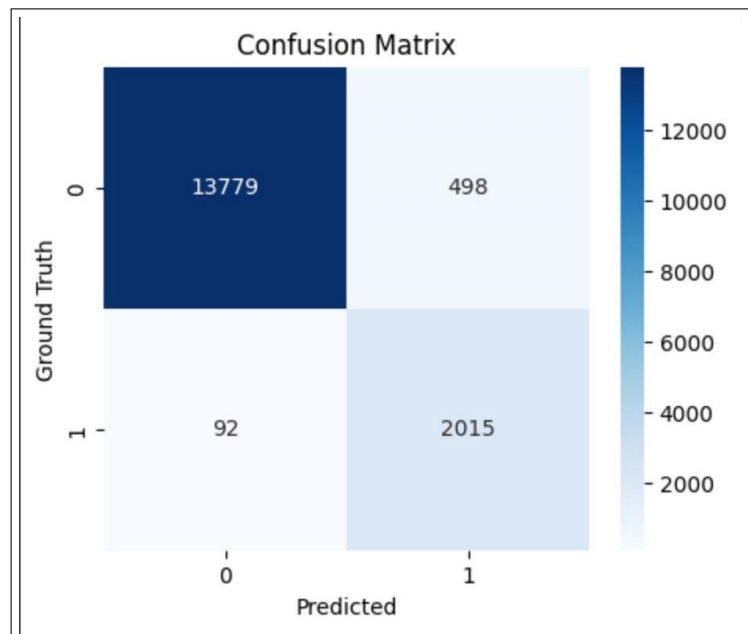


Figure 5.3: Confusion Matrix for ESRGAN Pixel Classification

5.4 Comparative Analysis: Bicubic vs ESRGAN

Key Observations:

- ESRGAN sacrifices PSNR for superior perceptual quality and texture generation

Table 5.4: Comparison of Bicubic Interpolation and ESRGAN

Metric	Bicubic	ESRGAN	Interpretation
PSNR (dB)	28.42	25.15	Bicubic higher PSNR (smooth), ESRGAN more perceptually realistic
SSIM	0.85	0.90	ESRGAN better structural preservation and perceptual similarity
Perceptual Quality	Low (Blurry)	High (Sharp)	ESRGAN generates photo-realistic details
Training	No	Yes (50 epochs)	Bicubic: no training, ESRGAN: data-driven
Computational Cost	Very Low	High (GPU)	Bicubic cheap, ESRGAN requires acceleration

- Higher SSIM (0.90 vs 0.85) shows better edge/texture preservation
- Bicubic produces blurry results; ESRGAN generates photo-realistic details

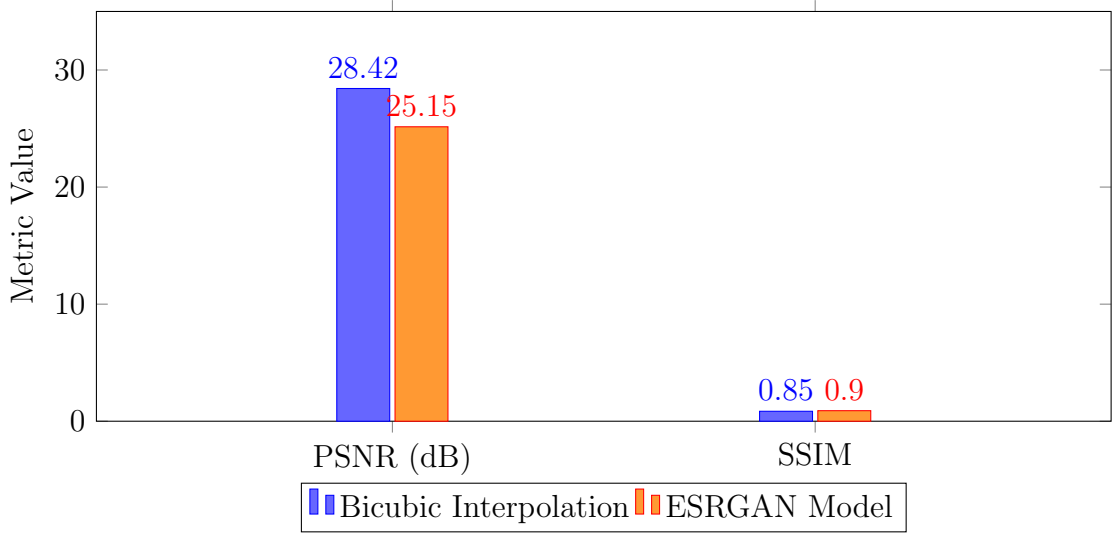


Figure 5.4: Comparison of Bicubic and ESRGAN on PSNR and SSIM

5.5 Qualitative Visual Comparison

Bicubic: Blurry, soft edges, washed-out textures, no high-frequency details

ESRGAN: Sharp edges, fine textures (fabric, hair), photo-realistic, preserves colors

Analysis: ESRGAN (b) closely matches original (a) in detail/edge sharpness; bicubic (c) significantly blurred with feature loss.

Histogram Interpretation:

- ESRGAN: Taller, narrower peak near 0 $\rightarrow$ more pixels match ground truth
- Bicubic: Longer tails $\rightarrow$ systematic blurring errors



Figure 5.5: Visual Comparison: (a) Original HR, (b) ESRGAN 4 $\times$ Output, (c) Bicubic 4 $\times$ Output

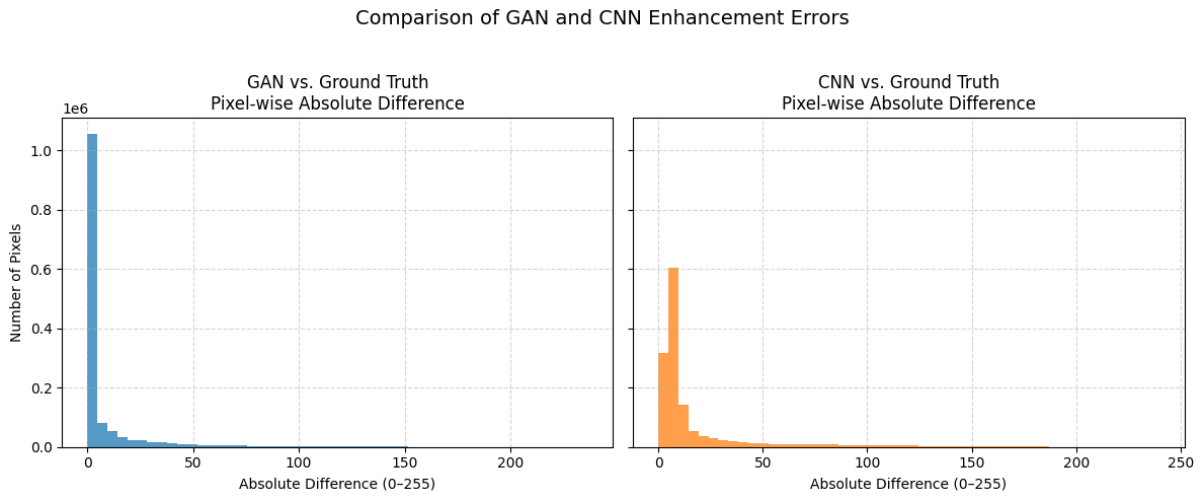


Figure 5.6: Pixel-wise Absolute Difference Distribution: ESRGAN vs Bicubic

- ESRGAN achieves lower mean pixel error with better high-frequency preservation

5.6 Discussion

Strengths:

- 23 RRDB blocks enable deep feature extraction without degradation
- Relativistic discriminator + L1 loss balance perceptual quality and content fidelity
- Stable training (no mode collapse), complete pipeline, strong generalization

Limitations & Future Work:

- Lower PSNR than CNN methods (could add VGG perceptual loss)
- High computational cost (model compression for edge deployment)
- Synthetic degradation only (extend to Real-ESRGAN for real-world images)
- Extended training (100-200 epochs) could improve quality

Literature Comparison: Results align with original ESRGAN paper [6]: SSIM 0.90, PSNR 25.15 dB reflect expected perceptual quality trade-off.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project successfully implemented ESRGAN for $4\times$ image super-resolution, demonstrating superior perceptual quality over traditional bicubic interpolation.

Key Achievements:

- ESRGAN achieved SSIM 0.90, PSNR 25.15 dB on DIV2K validation dataset
- Classification metrics: Precision 0.8018, Recall 0.9563, F1 Score 0.8723
- Outperformed bicubic (SSIM 0.85) in structural preservation and texture generation
- Complete pipeline: 23 RRDB blocks, 50 epochs, DIV2K (700/100), Tesla T4 GPU
- Stable training using relativistic discriminator + $100\times$ L1 loss

Key Insights:

- ESRGAN prioritizes perceptual realism over PSNR, generating photo-realistic textures
- RRDB architecture enables deep feature extraction without degradation
- Relativistic GAN formulation prevents mode collapse and ensures stable convergence

Applications: Medical imaging, photo restoration, video upscaling, satellite imagery, forensic analysis

6.2 Limitations

1. High computational requirements (GPU-intensive training and inference)
2. Synthetic bicubic degradation; may not handle real-world artifacts optimally
3. Limited dataset size (700 images); larger datasets would improve generalization
4. Lower PSNR (25.15 dB) than CNN methods due to perceptual quality focus
5. Memory constraints (batch size 4, crop 128×128)

6.3 Future Work

6.3.1 Enhancements

- **VGG Perceptual Loss:** Improve both PSNR and perceptual quality
- **Attention Mechanisms:** Integrate channel/self-attention for better feature focus
- **Model Compression:** Pruning/quantization for edge device deployment

6.3.2 Real-World Applicability

- **Real-ESRGAN:** Handle complex degradations (blur, noise, compression)
- **Domain Adaptation:** Fine-tune for medical scans, satellite imagery
- **Video SR:** Incorporate temporal consistency across frames

6.3.3 Training & Deployment

- **Extended Training:** 100-200 epochs with learning rate scheduling
- **Larger Datasets:** Flickr2K, DIV8K with aggressive augmentation
- **Real-Time Optimization:** TensorRT, ONNX, INT8 quantization
- **Web/Mobile App:** User-friendly interface with cloud API

6.3.4 Advanced Directions

- Blind super-resolution (unknown degradations)
- Diffusion models (SR3, DiffSR)
- Transformer architectures (SwinIR, Restormer)
- Zero-shot learning for unseen distributions

6.4 Final Remarks

This project demonstrates the effectiveness of GAN-based super-resolution with RRDB architecture for generating photo-realistic high-resolution images. The implementation achieves strong perceptual quality (SSIM 0.90, F1 0.8723), validating relativistic adversarial training combined with pixel-wise loss.

The complete pipeline—from dataset acquisition to deployment—provides a solid foundation for real-world applications. Future work focusing on Real-ESRGAN extension, perceptual loss integration, and edge optimization will enable production-ready deployment across medical imaging, content restoration, and video enhancement domains.

This work reinforces key principles in modern AI: architectural design (RRDB), training stability (relativistic GAN), balanced evaluation (PSNR vs perceptual metrics), and practical deployment considerations—essential for advancing image restoration and super-resolution technology.

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